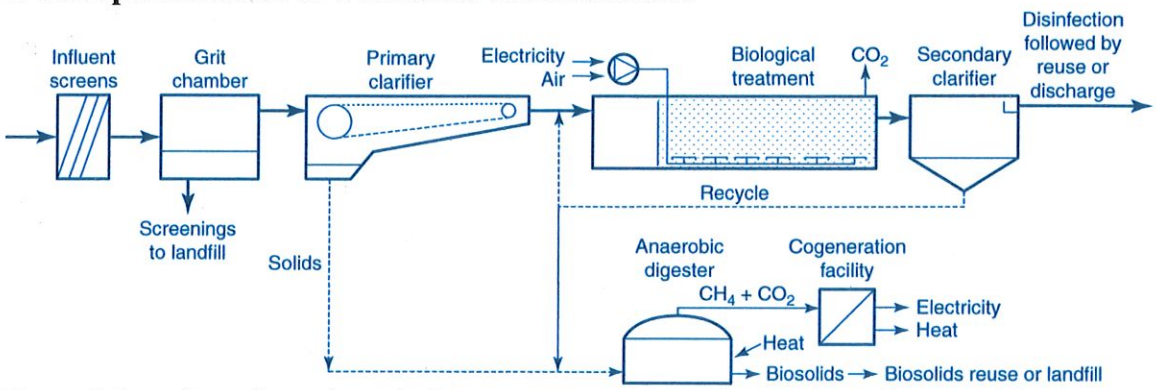
Indiana University
Fall 2020

SPEA-E 552
A. Ward

Name: SOLUTION GRADE NOTES

Quiz #4 (Lectures 15-18)

1. Conceptual function of Wastewater Treatment Plant



Pictured above is a schematic typical wastewater treatment plant. Below, please explain the purpose of each unit process in a few sentences. No equations allowed. Your answers should convey that you understand the purpose or function of each unit process:

1a: Aeration basin (labeled 'biological treatment' in the image above)

+4 POINTS EACH MAXIMUM

1b: Secondary Clarifier

ANSWERS WILL VARY

1c: Anaerobic Digester

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2. At Indiana University, the population of about 44,000 students (in a non-COVID year) generate wastewater at a rate of about 0.57 m^3 per person per day. Wastewater has a typical concentration of 240 g TSS/m^3. In this problem we will analyze the energy potential in this waste stream.

A. What is the mass rate of TSS arriving at the wastewater treatment plant each day? (kg TSS/d)

$$44{,}000 \text{ people} \times \frac{0.57\ m^3}{\text{person}\cdot d} \times \frac{240\ g}{m^3} \times \frac{1\ kg}{1000\ g} = \boxed{6019.2\ kg/d}$$

+2 APPROACH
+1 SOLUTION #
+1 UNITS

B. Using a quantitative flow diagram approach (Lecture 17), what is the mass rate arriving at the anaerobic digester each day? (kg TSS/d)

$$6019.2\ \frac{kg}{d} * \underbrace{(0.65 + 0.35 * 0.85 * 0.85)}_{\sim 0.903} = \boxed{5{,}434.6\ kg/d}$$

+2 APPROACH
+1 SOLUTION #
+1 UNITS

C. Assume 35% of the mass arriving at the anaerobic digester is converted to gas, and the produced gas contains 65% CH_4. What mass rate of CH_4 is produced? (kg/d)

$$5434.6\ \frac{kg}{d} * 0.35 * 0.65 = \boxed{1236.4\ \frac{kg}{d}}$$

+2 APPROACH
+1 SOLUTION #
+1 UNITS

D. Methane contains about 10.3 kW-hr per m^3 of total energy. Of this, about 25% can be recovered as electrical energy after combustion. What is the electrical energy that could be produced from digesting sludge generated by IU students? (kW-hr / d)

$$\frac{1236.4\ kg\ CH_4}{d} * \frac{1\ m^3\ CH_4}{0.8\ kg\ CH_4} \times \frac{10.3\ kW\cdot hr}{m^3} * 0.25 = \boxed{\frac{3980\ kW\cdot hr}{d}}$$

+2 APPROACH
+1 SOLUTION #
+1 UNITS

E. Methane contains about 39 MJ/m^3 of total energy. If this energy is released in the form of heat, what volume of water could be heated from 20 deg-C to 100 deg-C by burning the methane produced? (m^3/d)

$$\frac{1236.4\ kg\ CH_4}{d} * \frac{1\ m^3}{0.8\ kg} * \frac{39\ MJ}{m^3} \times \frac{10^6\ J}{1\ MJ} \times \frac{9\ °C}{4.186\ J} \times \frac{1}{80°C} \times \frac{1\ m^3}{1000\ kg}$$

$$\ldots \times \frac{1\ kg}{1000\ g} = \boxed{179.9\ \frac{m^3}{d}}$$

+2 APPROACH
+1 SOLUTION #
+1 UNITS

F. The electrical generation potential (part D) represents about 0.52% of IU's electrical use on a typical day. The heat generation (part E) represents about 2.6% of IU's peak demand for steam to heat and cool campus buildings. In your opinion, would it be logical for IU to invest in operating a WWTP to recover this energy for on-campus use? Why or why not? (no wrong answers, just provide your opinion and rationale).

ANSWERS WILL VARY

UP TO +4